

CHOCOLATE SALES ANALYSIS AND PROFIT PREDICTION USING MACHINE LEARNING

Project Report

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Introduction

The Chocolate Sales Analysis and Profit Prediction project aims to analyze a large-scale sales dataset using Python and Machine Learning techniques. The project focuses on understanding sales patterns, customer purchasing behaviour, product performance, and profit generation. Various visualization techniques such as Bar Charts, Line Charts, Pie Charts, Scatter Plots, and Histograms are used to gain meaningful business insights. A Decision Tree Regressor model is developed to predict profit based on sales-related features and evaluate predictive performance.

Objectives

- Analyze chocolate sales data using Python.
- Perform data preprocessing and feature engineering.
- Visualize important business metrics using charts.
- Identify profitable product categories.
- Study revenue trends over time.
- Build a machine learning model for profit prediction.
- Evaluate model performance using MAE and R^2 Score.

Dataset Description

The dataset contains sales transactions, product information, customer details, store information, and calendar data. These datasets were integrated to perform business analytics and predictive modelling.

Dataset	Records
Sales	960,935
Products	200
Customers	50,000
Stores	100
Calendar	731

Data Preprocessing

The raw datasets were loaded into Python using the Pandas library. The `order_date` column was converted into datetime format to enable time-based analysis. Additional features such as year, month, and month_name were extracted from the date field. Sales and product datasets were merged using the `product_id` key to enrich transaction records with product information. The data was checked for consistency and prepared for visualization and machine learning tasks.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the dataset. Key business metrics such as revenue, profit, product categories, and customer transactions were analyzed. Descriptive statistics and visualizations helped identify patterns, trends, and relationships within the data. EDA provided valuable insights that guided further analysis and model development.

Data Visualization Section

Bar Chart

Total Profit by Chocolate Category

The bar chart shows total profit generated by each chocolate category. Categories with taller bars contribute more profit to the business, helping identify the most profitable product segments.

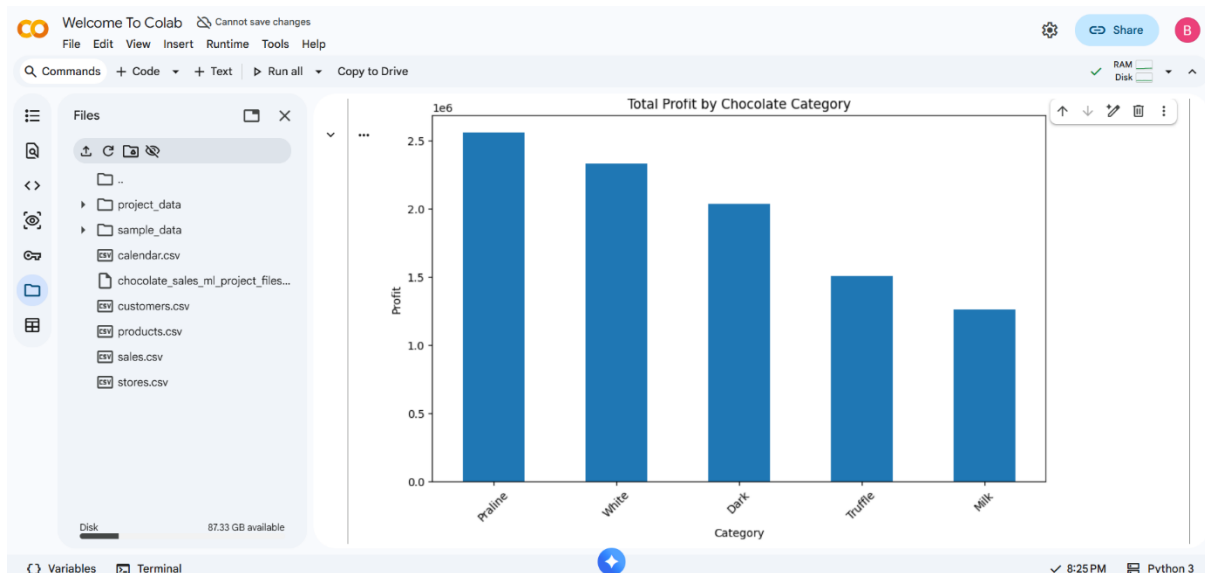


Figure 1: Total Profit by Chocolate Category

Line Chart

Monthly Revenue Trend

The line chart illustrates revenue fluctuations across different months. Peaks indicate high-sales periods, while dips indicate lower business activity. This trend analysis helps understand seasonal sales patterns.

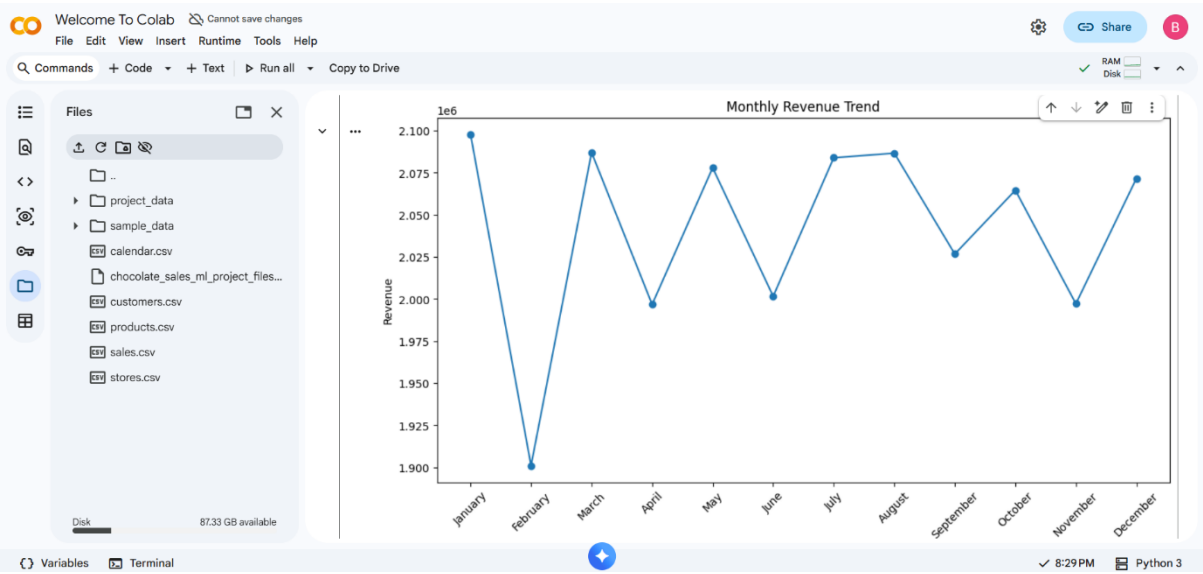


Figure 2: Monthly Revenue Trend

Pie Chart

Revenue Share by Chocolate Category

The pie chart represents the contribution of each chocolate category to total revenue. Larger slices indicate categories generating higher revenue and market demand.

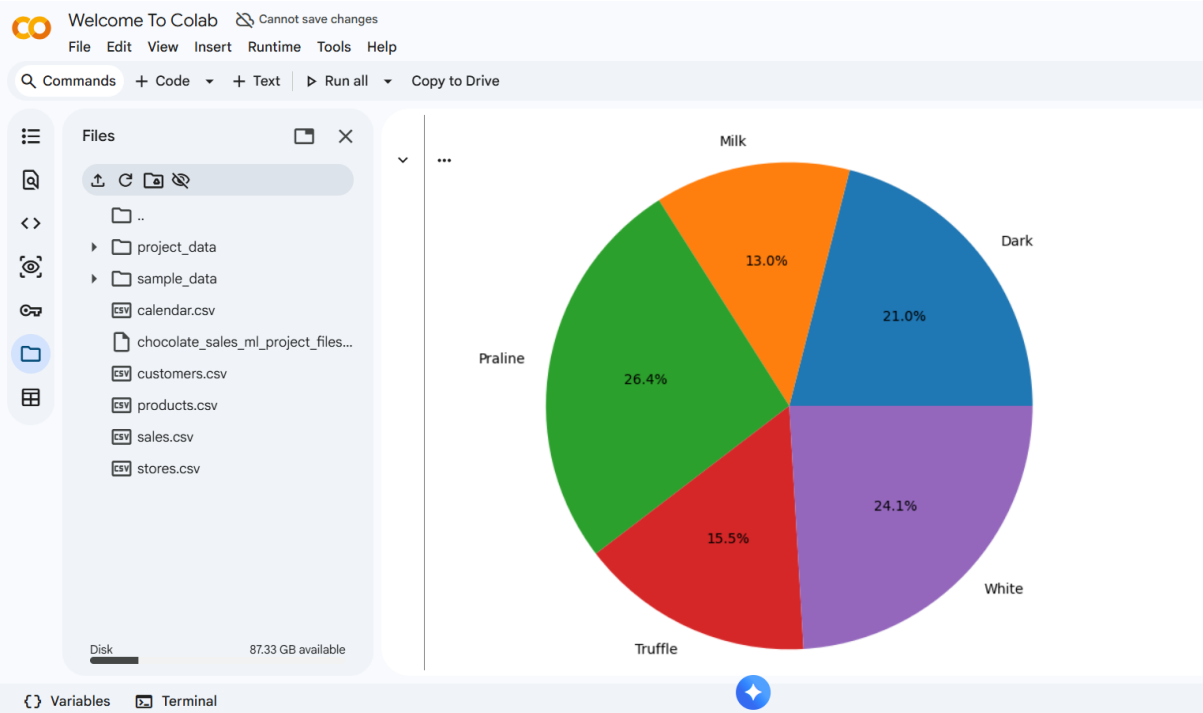


Figure 3: Revenue Share by Category

Scatter Plot

Revenue vs Profit

The scatter plot visualizes the relationship between revenue and profit. A positive trend indicates that higher revenue transactions generally generate higher profit.

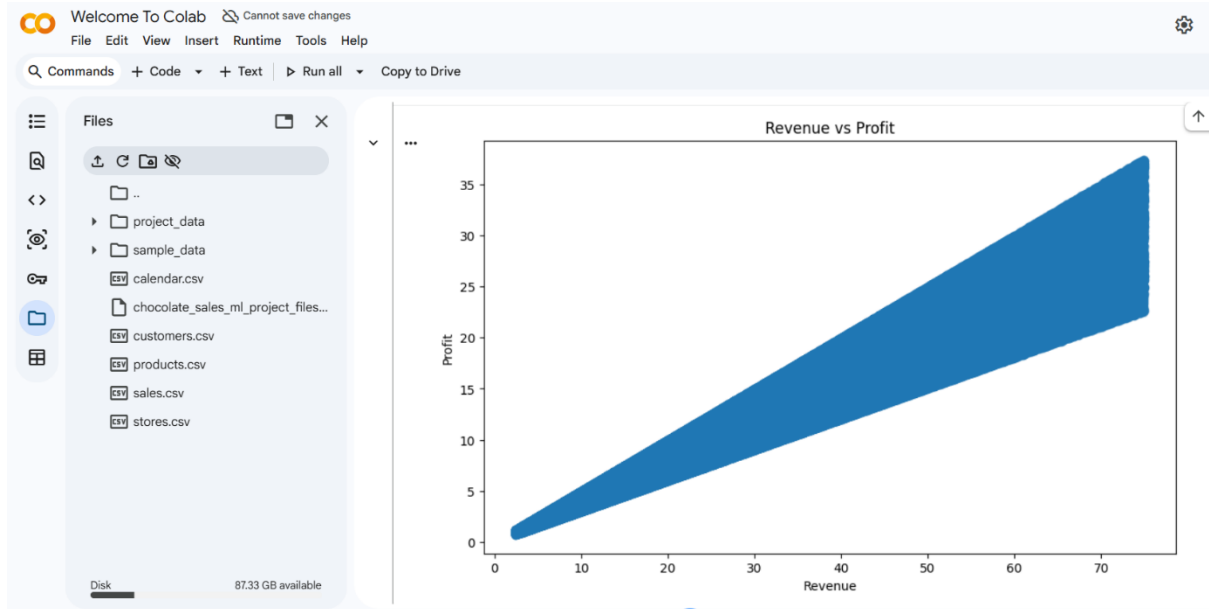


Figure 4: Revenue vs Profit Relationship

Histogram

Profit Distribution

The histogram shows how profit values are distributed across transactions. Most sales fall within the central profit range, while very high-profit transactions occur less frequently.

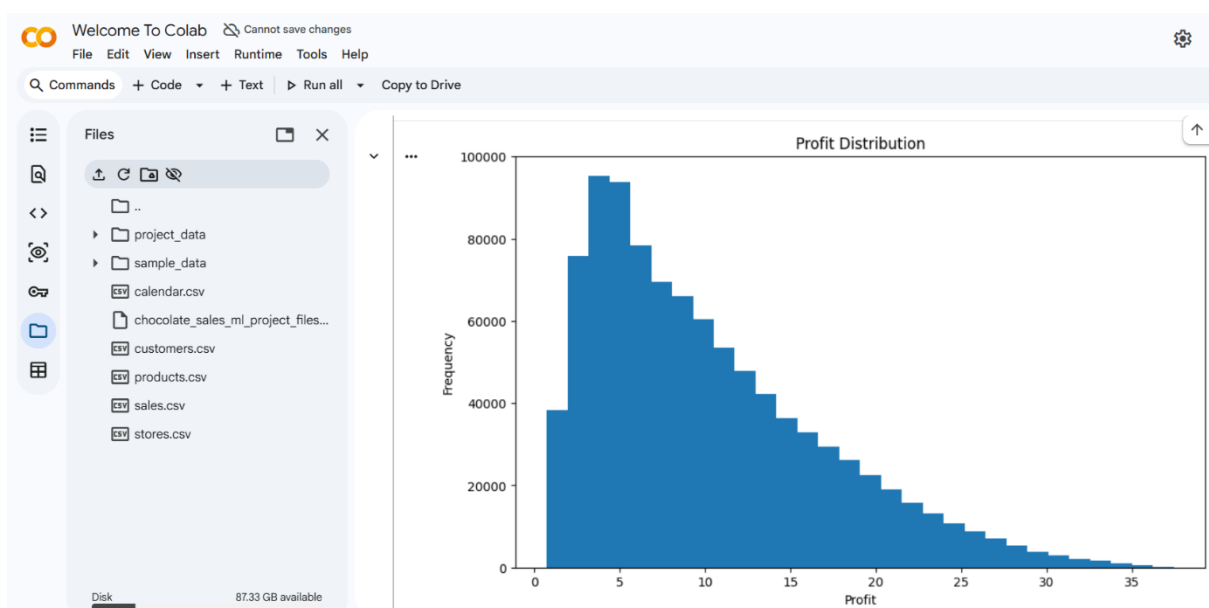


Figure 5: Profit Distribution

Machine Learning Model

A Decision Tree Regressor was used to predict profit based on sales-related features. The dataset was divided into training and testing sets using an 80:20 ratio. The model learned patterns from historical sales data and generated profit predictions for unseen records. Model performance was evaluated using Mean Absolute Error (MAE) and R^2 Score.

Results and Discussion

Metric	Value
MAE	0.0167
R^2 Score	0.99998

ML Report Interpretation

A Decision Tree Regressor model was trained to predict profit using sales-related features such as quantity, unit price, discount, revenue, and cost. The model performance was evaluated using Mean Absolute Error (MAE) and R^2 Score. Higher R^2 values indicate better prediction accuracy.

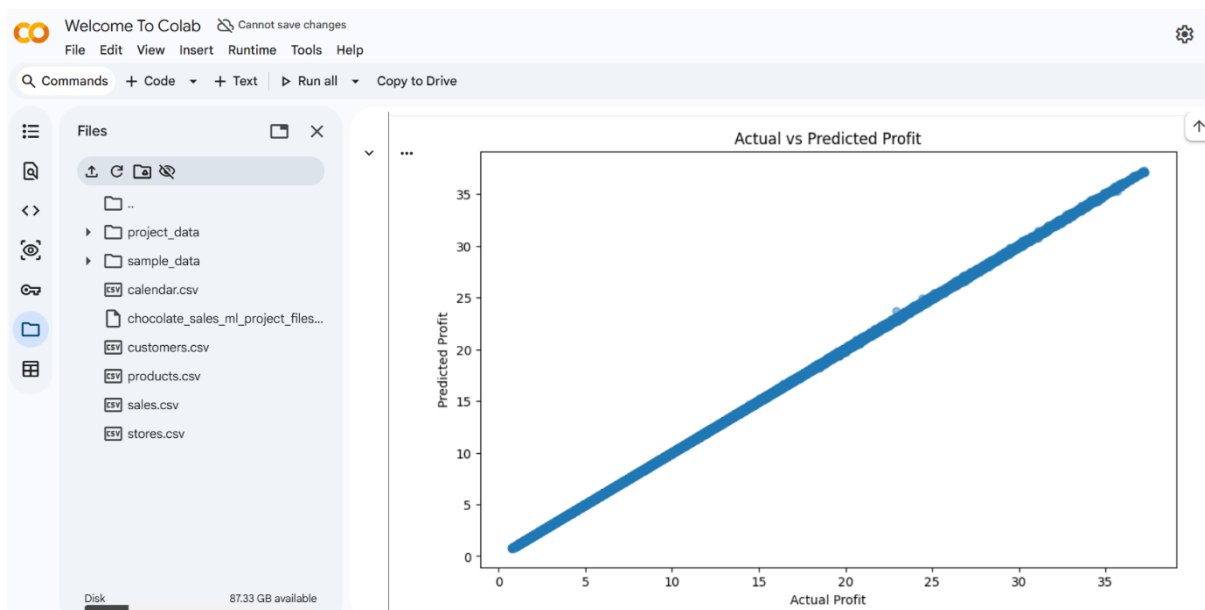


Figure 6: Model Results

Actual vs Predicted Profit

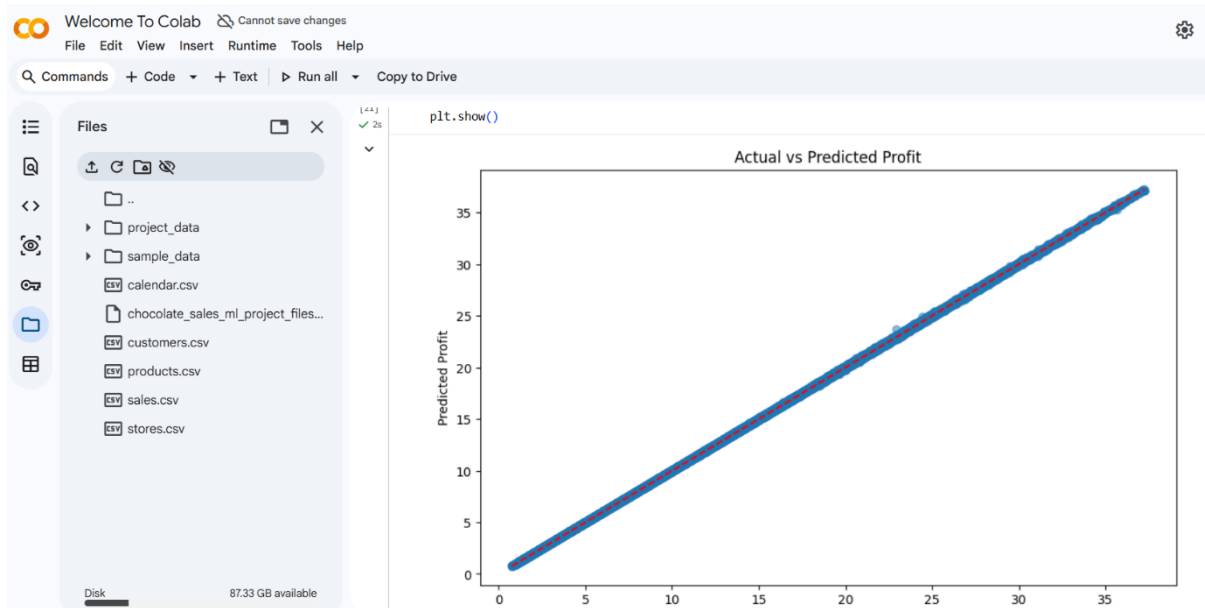


Figure 7: Actual vs Predicted Results

Conclusion

This project successfully analyzed chocolate sales data using Python, data analytics, and machine learning techniques. Various visualizations helped identify revenue trends, profit distribution, and category performance. A Decision Tree Regressor model was developed to predict profit with high accuracy. The project demonstrates how data-driven approaches can support business decision-making and improve understanding of sales performance.

Future Scope

Future enhancements may include the use of advanced machine learning algorithms such as Random Forest, XGBoost, or Neural Networks for improved predictive performance. Additional business intelligence dashboards can be developed using Power BI or Tableau. Real-time sales monitoring and forecasting systems can also be implemented for practical business applications.

References

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